

Database Schema

Management approach

Mechanism	Role
Hibernate <code>ddl-auto=update</code>	Creates/updates most tables from entities
<code>schema.sql</code>	Creates <code>account_ordinal_seq</code>
Seed runners	<code>RoleInitializer</code> , <code>AdminRoleInitializer</code> , <code>AccountPolicyInitializer</code>

Entity-relationship (current)

```
erDiagram
    customers ||--o{ account : has
    customers ||--o{ beneficiary : saves
    customers ||--o{ transfer_request : requests
    users ||--o{ user_roles : has
    roles ||--o{ user_roles : grants
    account ||--o{ bank_transaction : posts
    account_policy {
        bigint policy_id PK
        string account_type
        string currency_code
    }
    customers {
        bigint customer_id PK
        decimal transfer_approval_threshold
    }
    account {
        bigint account_id PK
        string account_number UK
        bigint customer_id FK
    }
    users {
        bigint user_id PK
        string username UK
        bigint customer_id
    }
    beneficiary {
        bigint beneficiary_id PK
        bigint customer_id FK
        string bank_type
    }
    transfer_request {
        bigint transfer_request_id PK
        bigint customer_id FK
        string status
    }
    audit_event {
        bigint id PK
        string category
        string action
    }
    bank_transaction {
        bigint transaction_id PK
        bigint account_id FK
    }
```

Tables

customers

Entity: `com.bankone.customer.entity.Customer`

Column	Notes
customer_id	PK, identity
first_name, last_name	Required
email, phone_number	Unique
date_of_birth	Optional
address, status	Required
transfer_approval_threshold	Optional; portal same-bank gate
created_at, updated_at	@PrePersist / @PreUpdate

customerCode is **not** stored; JSON accessor formats ID via BusinessIdFormatter.

account

Entity: com.bankone.account.entity.Account

Column	Notes
account_id	PK
account_number	Unique, generated
branch_code	String (no branch table)
account_type, currency_code, ordinal, check_digit	Number composition
available_balance, ledger_balance	Updated on open/deposit
debit_count, credit_count, last*_at	Counters / stamps
status	ACTIVE, FROZEN, DORMANT, SUSPENDED, CLOSED
customer_id	FK → customers
Audit-ish	created_at, activated_at, closed_at, created_by, closed_by

account_policy

Entity: com.bankone.account.entity.AccountPolicy extends AuditableEntity

Unique (account_type, currency_code).

Seeded INR policies for CURRENT, SAVINGS, SALARY, FIXED_DEPOSIT, RECURRING_DEPOSIT. **No LOAN seed.**

users, roles, user_roles, role_access

Entities under com.bankone.user / com.bankone.role. Sequences: user_seq, role_seq, user_role_seq.

`users.customer_id` — null for staff; set for portal logins.

`role_access` — element collection of access codes (`AppAccess`).

`AuditableEntity` columns on User/Role/UserRole/AccountPolicy: `created_at` , `updated_at` , `created_by` , `updated_by` , `version` .

`bank_transaction`

Entity: `com.bankone.transaction.entity.Transaction`

Table name is `bank_transaction` (avoids SQL keyword `transaction`).

Column	Notes
<code>transaction_id</code>	PK (identity)
<code>account_id</code>	FK → <code>account</code> (required)
<code>transaction_type</code>	Enum string: CREDIT, DEBIT
<code>amount</code>	> 0; precision 19,2
<code>balance_after</code>	Account ledger balance after posting
<code>currency_code</code>	Copied from account (length 3)
<code>narration</code>	Optional text
<code>created_at</code>	Set in <code>@PrePersist</code>
<code>created_by</code>	Staff username when provided

Written from open/deposit/withdraw/transfer via `TransactionService.record` . Staff list: `GET /transactions` .

`beneficiary`

Portal payees (`SAME_BANK` / `OTHER_BANK`), soft-deactivate via `active=false` .

`transfer_request`

Portal transfers awaiting or after staff resolution (`PENDING` , `APPROVED` , `REJECTED` , `EXECUTED`).

`audit_event`

Activity trail — see [MODULES/Audit.md](#).

Sequences

```
-- schema.sql
CREATE SEQUENCE IF NOT EXISTS account_ordinal_seq
START WITH 1 INCREMENT BY 1;
```

Used by `AccountRepository.getNextOrdinal()` . Also Hibernate sequences for users/roles/transfer_request/beneficiary.

Not present (planned)

No tables yet for: `branch` master, `loan` product, outbox (learning plan), rate-limit store (Redis), shard metadata.

Migration notes

Changing entity fields with `ddl-auto=update` alters tables in place. For production-ready history, replace with Flyway/Liquibase (see `EXTENSION_GUIDE` and `TECH_LEARNING_PLAN`).